PRODUCT SPECIFICATION

Product Name	**S68F** LoRa Transceiver Module
Version	**E**
Doc No	**901-12401EN**
Date	**2026/09/03**

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AcSiP Technology Corp.

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Document History

Date	Revised Contents	Revised By	Version
2020/02/10	Initial Version	PW	A
2020/11/16	Modify electrical characteristics, block diagrams and mechanical dimensions	Kenny	B
2022/12/22	Modify the packing information drawing and pin description	Kenny	C
2025/09/05	Modify the SMT preparation content	Kenny	D
2026/09/03	Modify module thickness tolerance and tray dimension	Kenny	E

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1. Description

The AcSiP S68F transceiver features the LoRa™ long range modem that provides ultra-long range spread spectrum communication and high interference immunity whilst minimizing current consumption.

S68F can achieve a sensitivity of over -138dBm using a TCXO and the related bill of materials. The high sensitivity combined with the integrated +22 dBm(max.) power amplifier yields an industry- leading link budget making it optimal for any application requiring range or robustness. LoRa™ also provides significant advantages in both blocking and selectivity over conventional modulation techniques, solving the traditional design compromise between range, interference immunity, and energy consumption.

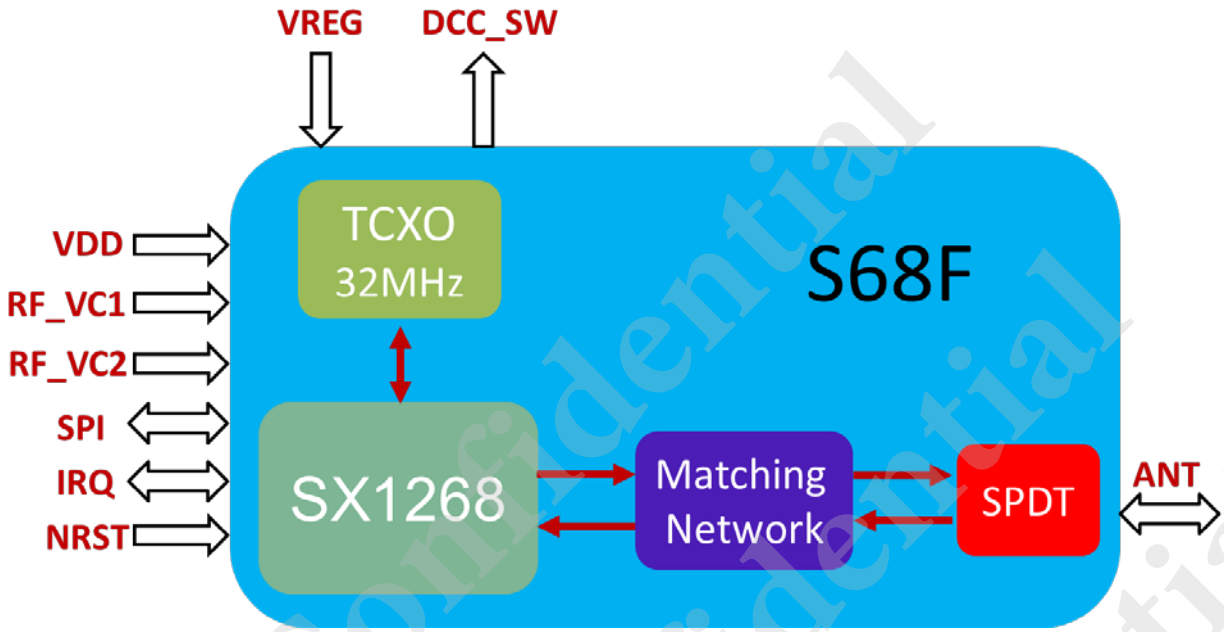
Feature

- Small size：9 mm x 8 mm x 1.26 mm (Typ.)
- LoRa Transceiver Module, +22 dBm Max. RF output power
- Programmable bit rate up to 62.5 kbps LoRa
- Programmable bit rate up to 300 kbps FSK
- High sensitivity: down to -138 dBm (@125KHz, SF12)
- Embedded TCXO of 32MHz

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1-1. Block Diagram

A simplified block diagram of the S68F module is depicted in the figure below.



1-2. Product Version

Part Number	Frequency Range	Spreading Factor	Bandwidth	Sensitivity (dBm)
S68F	433MHz - 510MHz	5 - 12	7.8 - 500 KHz	- 138 @125KHz, SF12

1-3. Specification

Technical Specifications	
Model Name	S68F
Product Description	LoRa Transceiver Module
Network Standard	Suitable PHY for LoRaWAN
Host Interface	SPI
Operation Conditions	
Temperature	■ Storage：-50℃ ~ +105℃ ■ Operating：-40℃ ~ +85℃
Humidity	■ Operating：10 ~ 95% (Non-Condensing) ■ Storage：5 ~ 95% (Non-Condensing)
Dimension	9 mm x 8 mm x 1.26 mm (Typ.)
Package	LGA type

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2. Electrical Characteristics

2-1. Absolute Maximum Ratings

Symbol	Parameter	Min.	Typ.	Max.	Unit
VDD	Supply Voltage	-0.5		3.9	V
Pmr	RF Input Level			+10	dBm

2-2. Recommended Operating Range

Symbol	Parameter	Min.	Typ.	Max.	Unit
VDD	Supply Voltage	1.8	3.3	3.7	V
Top	Temperature under bias	-40		85	°C

Note: VDD 3.3V for +22dBm , VDD 2.7 V for +20dBm , VDD 2.4 V for +19dBm

2-3. Power Consumption Specification

Symbol	Parameter	Conditions	Typ.	Max.	Unit
IDDSL	Supply current in Sleep mode		0.55	0.6	µA
IDDSBR	Supply current in Standby RC mode	TCXO OFF	0.58		mA
IDDSBT	Supply current in Standby XOSC mode	TCXO ON	1.9		mA
IDDR	Supply current in Receive mode	RX Boosted LoRa 125kHz	5.7		mA
IDDT	Supply current in Transmit mode with impedance matching	RF SetPW = + 22 dBm RF SetPW = + 20 dBm RF SetPW = + 17 dBm RF SetPW = + 14 dBm	112 99 81 65	115	mA

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2-4. RF Characteristics

2-4.1. Electrical Specifications

The electrical specifications are given with the following conditions unless otherwise specified:

- VDD = 3.3 V
- Temperature = 25 °C
- FRF = 433/510 MHz
- All RF impedances matched
- Transmit mode output power defined into a 50 ohm load impedance
- FSK BER = 0.1%, 2-level FSK modulation without pre-filtering, BR = 4.8 kb/s, FDA = ± 5 kHz, BW_F = 20 kHz double-sided
- LoRa® PER = 1%, packet 64 bytes, preamble 8 symbols, CR = 4/5, CRC on payload enabled
- RX/TX specifications given using default RX gain step and direct tie connection between Rx and Tx

2-4.2. Receive Mode Specifications

Symbol	Description	Conditions	Min	Typ	Max	Unit
RXS_2FB	Sensitivity 2-FSK, RX Boosted gain, split RF paths for Rx and Tx, RF switch insertion loss excluded	BR = 0.6 kb/s, FDA = 0.8 kHz, BW = 4 kHz BR = 1.2 kb/s, FDA = 5 kHz, BW = 20 kHz BR = 4.8 kb/s, FDA = 5 kHz, BW = 20 kHz BR = 38.4 kb/s, FDA = 40 kHz, BW = 160 kHz BR = 250 kb/s, FDA = 125 kHz, BW = 500 kHz		-125 -123 -118 -109 -104		dBm
RXS_LB	Sensitivity LoRa, Rx Boosted gain , split RF paths for Rx and Tx, RF switch insertion loss excluded	BW = 125 kHz, SF = 7 BW = 125 kHz, SF = 12 BW = 250 kHz, SF = 7 BW = 250 kHz, SF = 12 BW = 500 kHz, SF = 7 BW = 500 kHz, SF = 12		-124 -137 -121 -134 -117 -129		dBm
CCR_F	Co-channel rejection, FSK			-9		dB
CCR_L	Co-channel rejection, LoRa	SF = 7 SF = 12		5 19		dB
ACR_F	Adjacent channel rejection, FSK	Offset = ±50 kHz		45		dB
ACR_L	Adjacent channel rejection, LoRa	Offset = ±1.5 x BW, BW = 125 kHz, SF = 7 BW = 125 kHz, SF = 12		 60 72		dB

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Symbol	Description	Conditions	Min	Typ	Max	Unit
BI_F	Blocking immunity, FSK	BR = 4.8 kb/s, FDA = 5 kHz, BW = 20 kHz Offset = +/- 1 MHz Offset = +/- 2 MHz Offset = +/- 10 MHz		 68 70 80		dB
BI_L	Blocking immunity, LoRa	BW =125 kHz, SF = 12 Offset = +/- 1 MHz Offset = +/- 2 MHz Offset = +/- 10 MHz		 88 90 99		dB
FERR_L	Maximum tolerated frequency offset between transmitter and receiver, no sensitivity degradation, SF5 to SF12	All bandwidths, ±25% of BW The tighter limit applies (see below)		± 25%		BW
	Maximum tolerated frequency offset between transmitter and receiver, no sensitivity degradation, SF10 to SF12	SF12 SF11 SF10	-50 -100 -200		50 100 200	ppm

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2-4.3. Transmit Mode Specifications

Symbol	Description	Conditions	Min	Typ	Max	Unit
TXOP	Maximum RF output power	Highest power step setting		+21	+22	dBm
TXDRP	RF output power drop versus supply voltage	+22 dBm, VDD = 2.7 V +22 dBm, VDD = 2.4 V +22 dBm, VDD = 1.8 V		2 3 6		dB
TXPRNG	RF output power range	Programmable in 31 steps, typical value	TXOP-31		TXOP	dBm
TXACC	RF output power step accuracy			± 2		dB
TXRMP	Power amplifier ramping time	Programmable	10		3400	µs
TS_TX	Tx wake-up time	Frequency Synthesizer enabled @SF = 7		* 5 + PA ramping		ms

* (5ms + PA ramping) is verified under the condition of SF=7, the length of time will be fine-tuned by different SF conditions

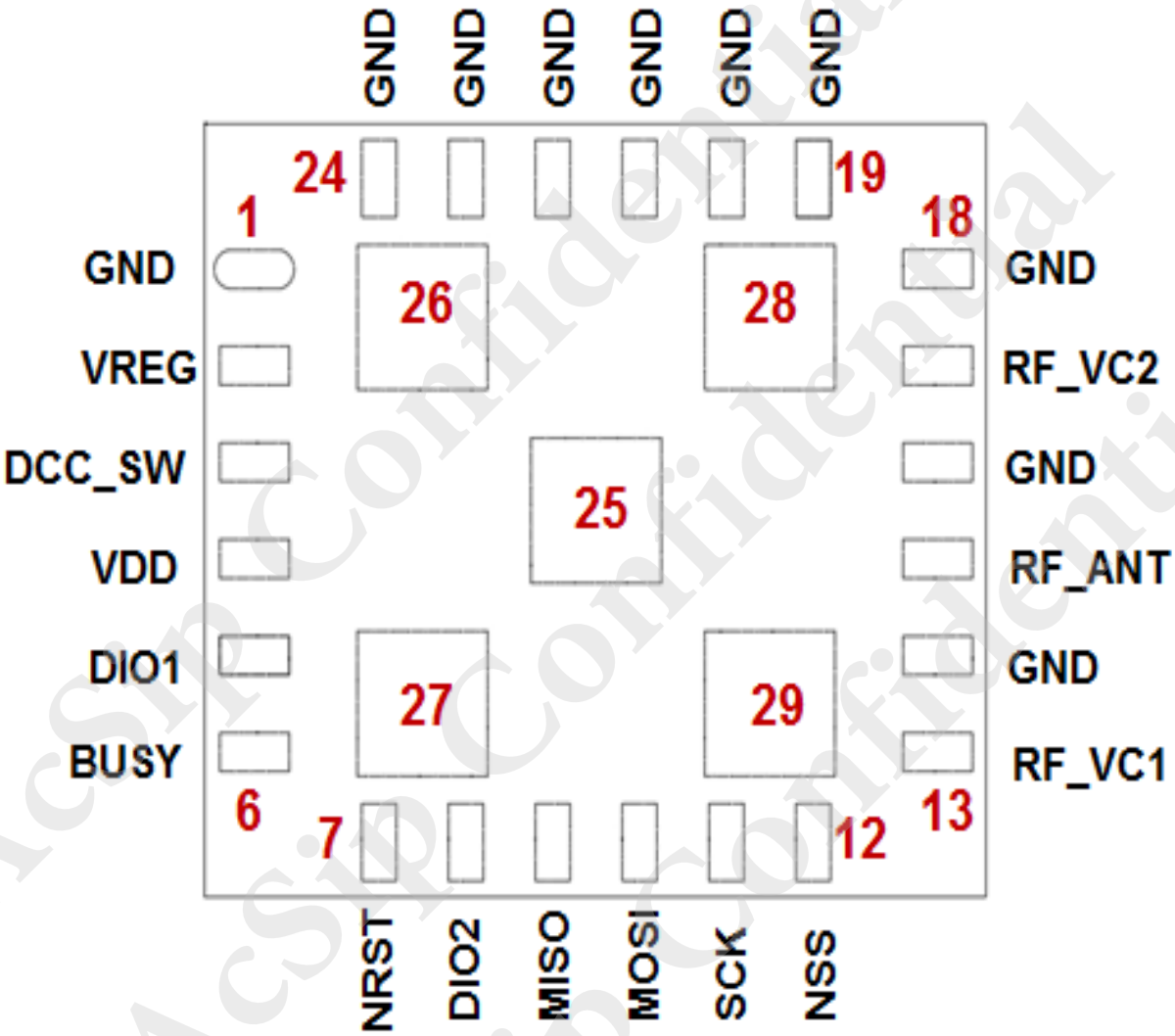
2-4.4. Digital Specification

Symbol	Description	Conditions	Min	Typ	Max	Unit
V_{IH}	Input High Voltage		0.7*VDD		VDD+0.3	V
V_{IL}	Input Low Voltage		-0.3		0.3*VDD	V
V_{IL-N}	Input Low Voltage for pin NRESET		-0.3		0.2*VDD	V
V_{OH}	Output High Voltage	Imax = -2.5 mA	0.9*VDD		VDD	V
V_{OL}	Digital output level low	Imax = 2.5 mA	0		0.1*VDD	V
Ileak	Digital input leakage current (NSS, MOSI, SCK)		-1		1	µA

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3. Pin Definition

3-1. Pin Description



Top_View

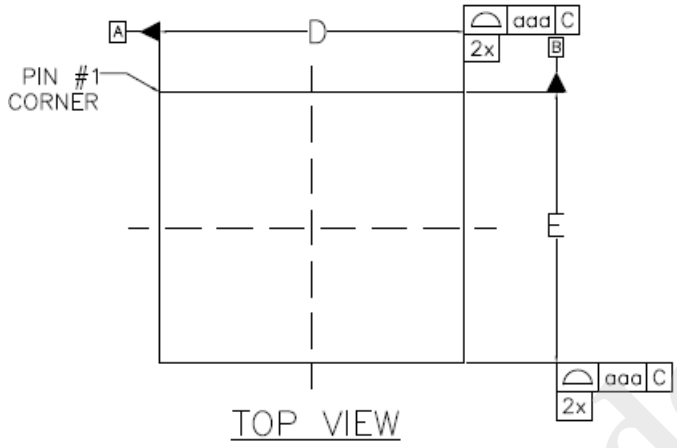
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Pin	Definition	I/O	Description
1	GND	-	Ground
2	VREG	I	Regulated voltage from the internal DC-DC
3	DCC_SW	O	DC-DC Switcher Output
4	VDD	I	Power supply
5	DIO1	I/O	Multi-purpose digital IO
6	BUSY	O	Busy indicator
7	NRST	I	Reset signal, active low
8	DIO2	I/O	* Multi-purpose digital IO / RF Switch control
9	MISO	O	SPI slave output
10	MOSI	I	SPI slave input
11	SCK	I	SPI clock
12	NSS	I	SPI Slave Select
13	RF_VC1	I	RF Switch control / (Pull “High” Transmitter ; Pull “Low” Receiver)
14	GND	-	Ground
15	RF_ANT	I/O	RF Input / Output
16	GND	-	Ground
17	RF_VC2	I	RF Switch control / (Pull “High” Receiver ; Pull “Low” Transmitter)
18	GND	-	Ground
19	GND	-	Ground
20	GND	-	Ground
21	GND	-	Ground
22	GND	-	Ground
23	GND	-	Ground
24	GND	-	Ground
25~29	EGND	-	Ground

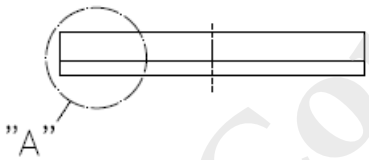
*DIO2 has a double functionality. As DIO2 can be used as a generic IRQ line and any IRQ can be routed through this pin. Also, DIO2 can be configured to drive an RF switch through the use of the command SetDio2AsRfSwitchCtrl(...). In this mode, DIO2 will be at a logical 1 during Tx and at a logical 0 in any other mode.

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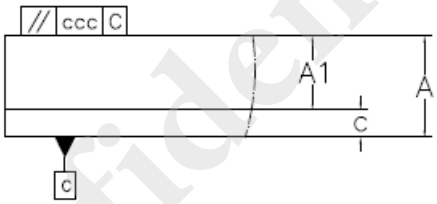
3-2. Mechanical Dimension



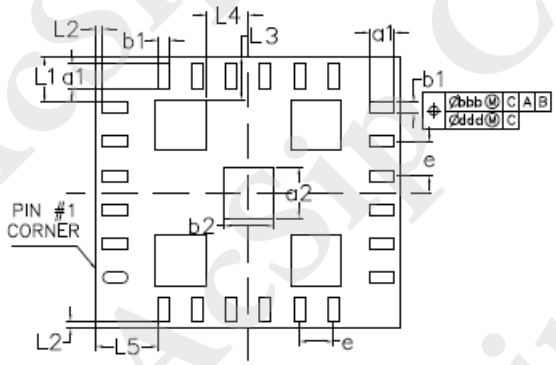
TOP VIEW



SIDE VIEW



DETAIL : "A"

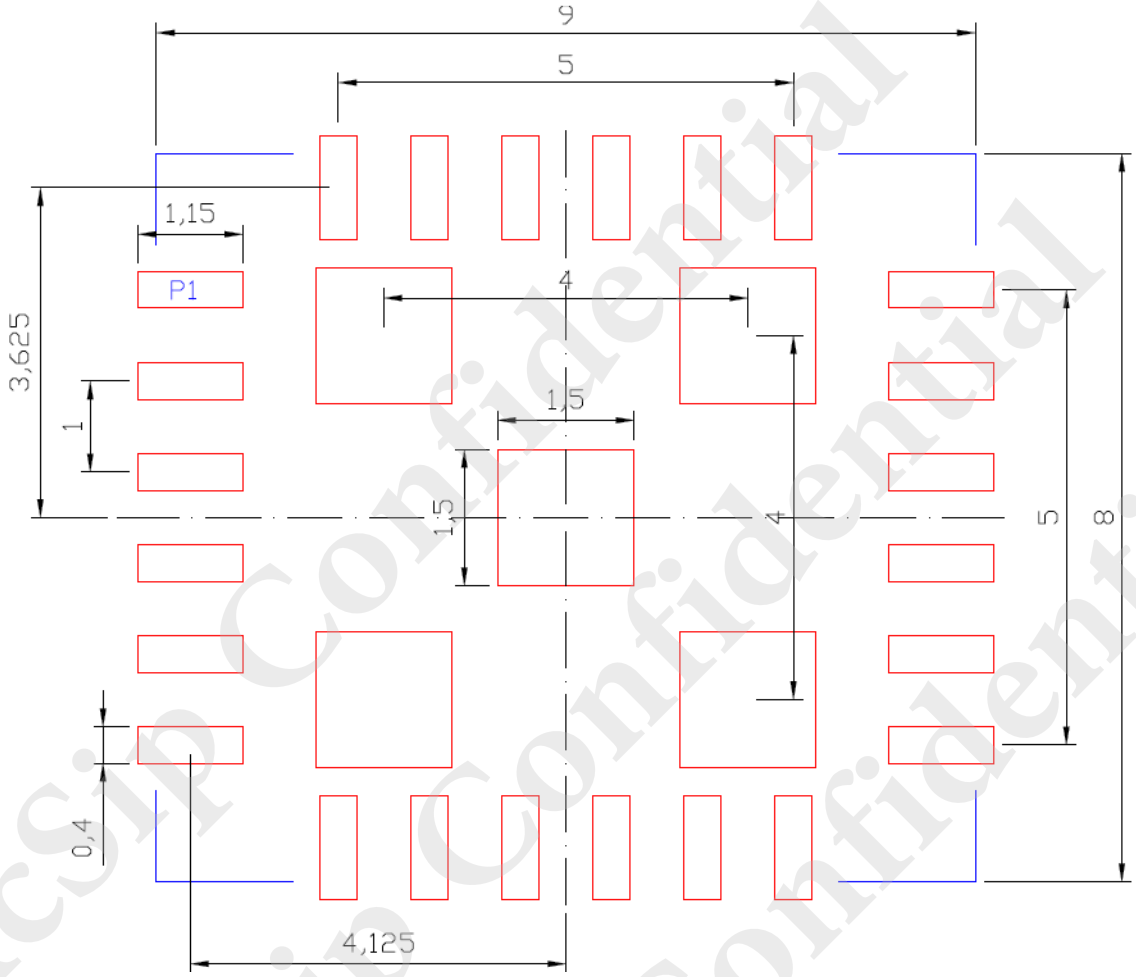


BOTTOM VIEW

Symbol	Dimension in mm		
	MIN	NOM	MAX
A	1.16	1.26	1.36
c	0.36	0.41	0.44
A1	0.83	0.85	0.87
D	8.90	9.00	9.10
E	7.90	8.00	8.10
a1		0.73	
b1		0.33	
a2		1.50	
b2		1.50	
e		1.00	
L1		1.335	
L2		0.20	
L3		1.25	
L4		1.25	
L5		1.835	
aaa		0.15 BSC	
bbb		0.10 BSC	
ccc		0.10 BSC	
ddd		0.05 BSC	

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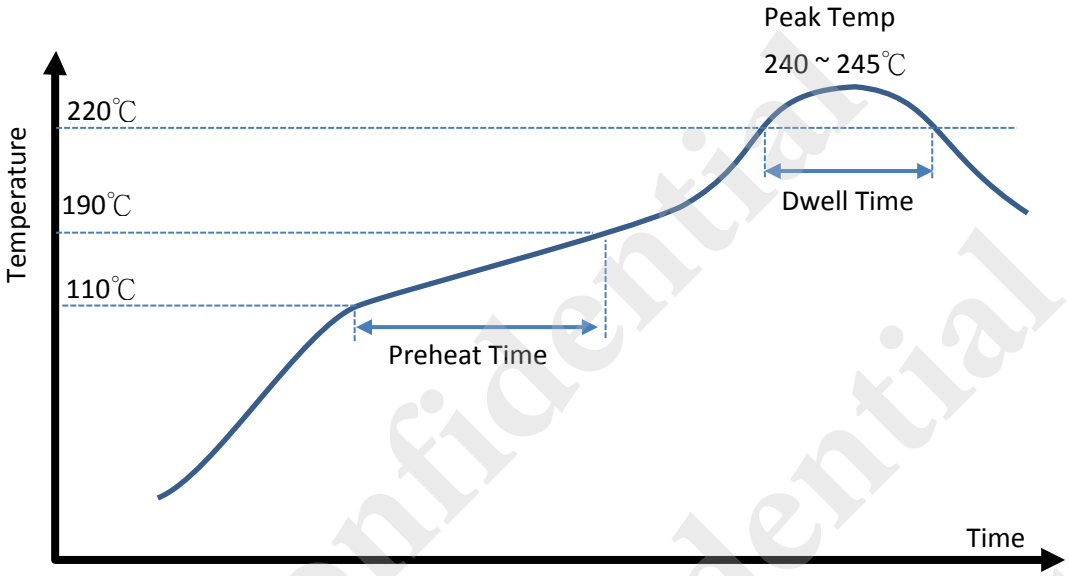
3-3. Recommended Footprint



Unit: mm

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4. Recommended Reflow Profile



Preheat time	110 ~ 190 ℃: 90 ~ 120 sec
Dwell time	above 220 ℃: 50 ~ 70 sec
Peak Temp	240 ~ 245 ℃
Ramp Up/Down Rate	Up : 1~3 ℃/sec Down: 1~5 ℃/sec

The recommended reflow profile is provided as a guideline. Optimal profile may differ due to oven type, assembly layout or other process variables. Nitrogen atmosphere is strongly recommended for best soldering result.

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5. Module Preparation

5-1. Handling

Handling the module, operator must wear the anti-static wrist strap to avoid ESD damage. After each module was aligned and tested, it should be transported and stored with anti -static tray and packing. This protective package must be remained in the suitable environment until the module is assembled and soldered onto the main board.

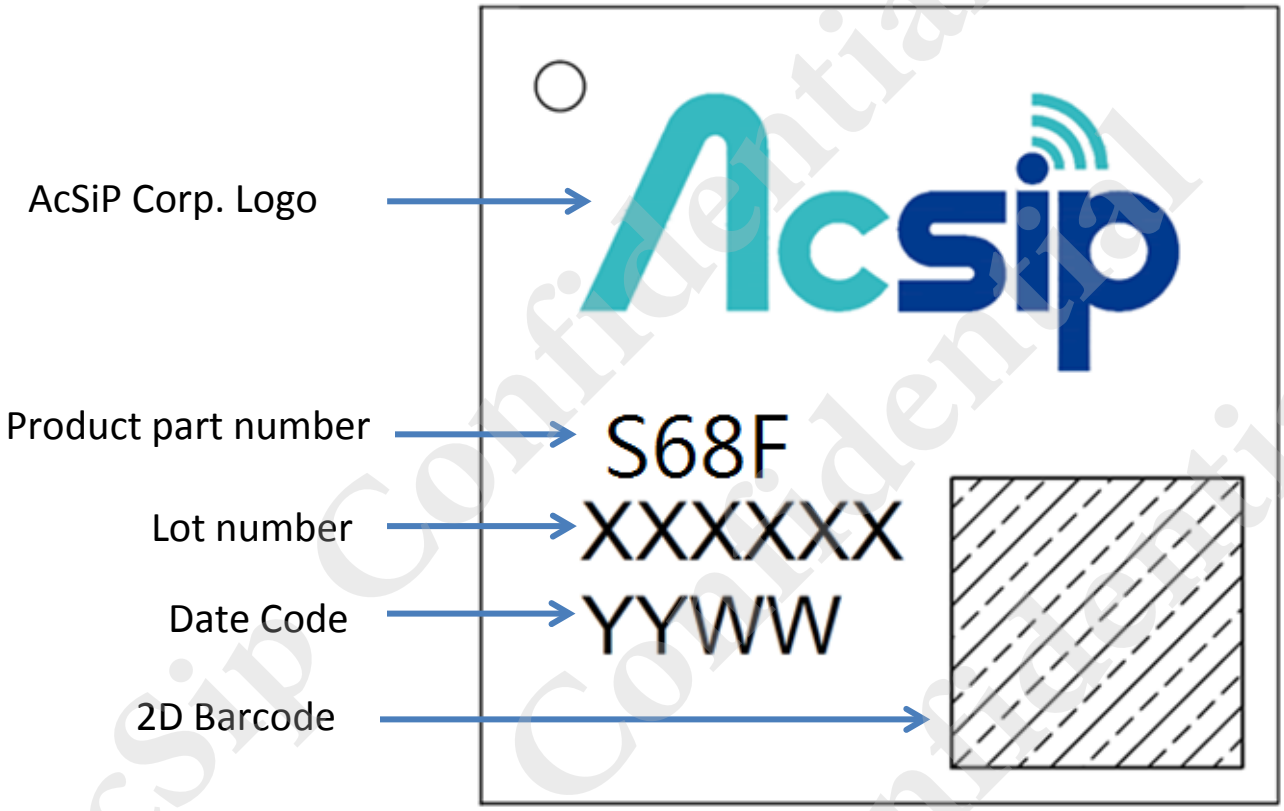
5-2. SMT Preparation

1. Calculated shelf life in sealed bag: 6 months at < 40℃ and < 90% relative humidity (RH).
2. Peak package body temperature: 250℃.
3. Moisture sensitivity level: 3
4. After bag was opened, devices that will be subjected to reflow solder or other high temperature process must.
   A. Mounted within: 168 hours of factory conditions < 30℃/60%RH.
   B. Stored at ≦ 10%RH with N2 flow box.
5. Devices require baking, before mounting, if:
   A. Package bag does not keep in vacuumed while first time open.
   B. Humidity Indicator Card is > 10% when read at 23±5℃.
   C. Expose at 5A condition over 8 hours or Expose at 5B condition over 24 hours.
6. If baking is required, devices may be baked for 12 hours at 125±5℃
7. Reflow condition: Please refer to IPC / JEDEC-J-STD-033

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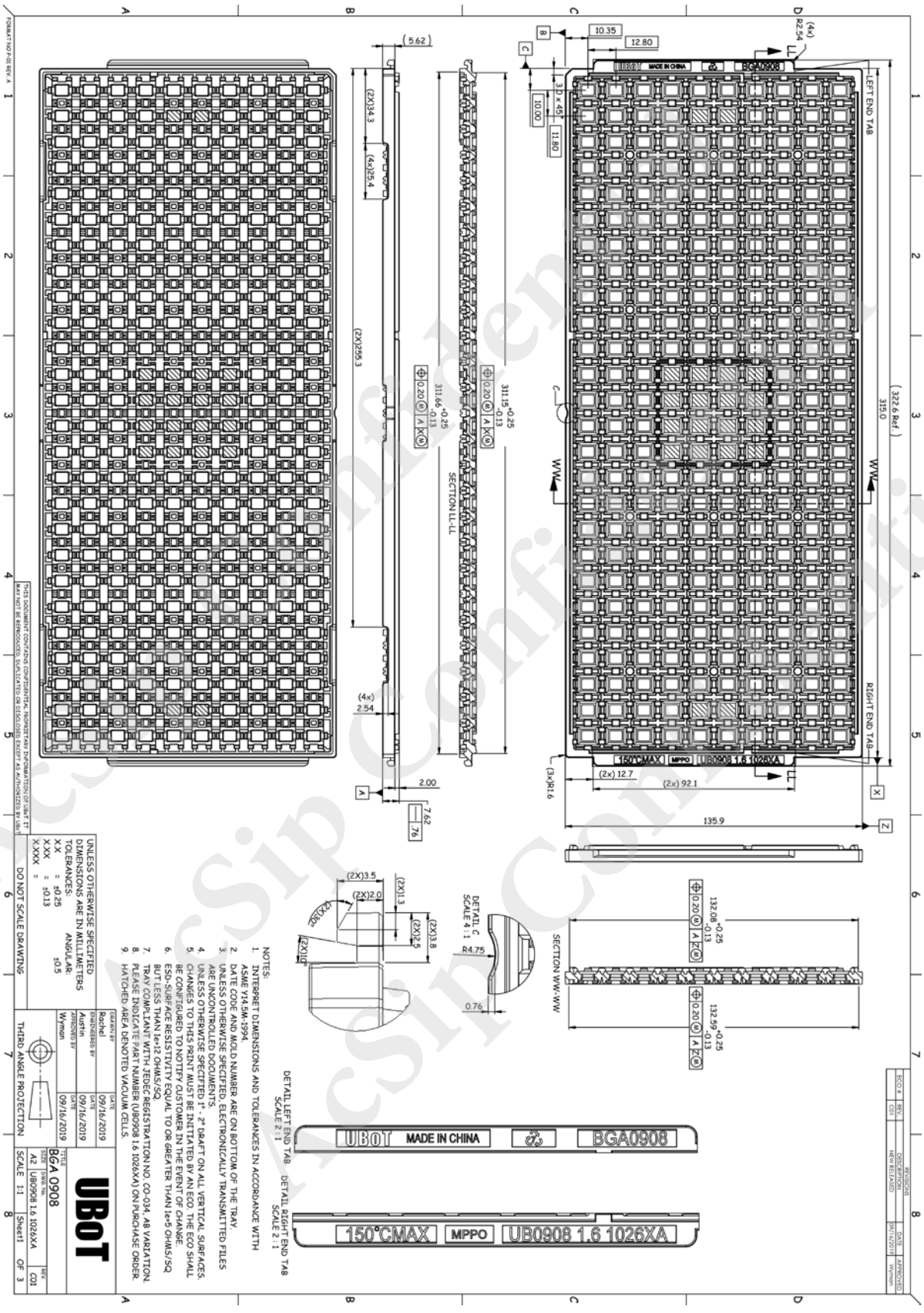
6. Package Information

6-1. Product Marking



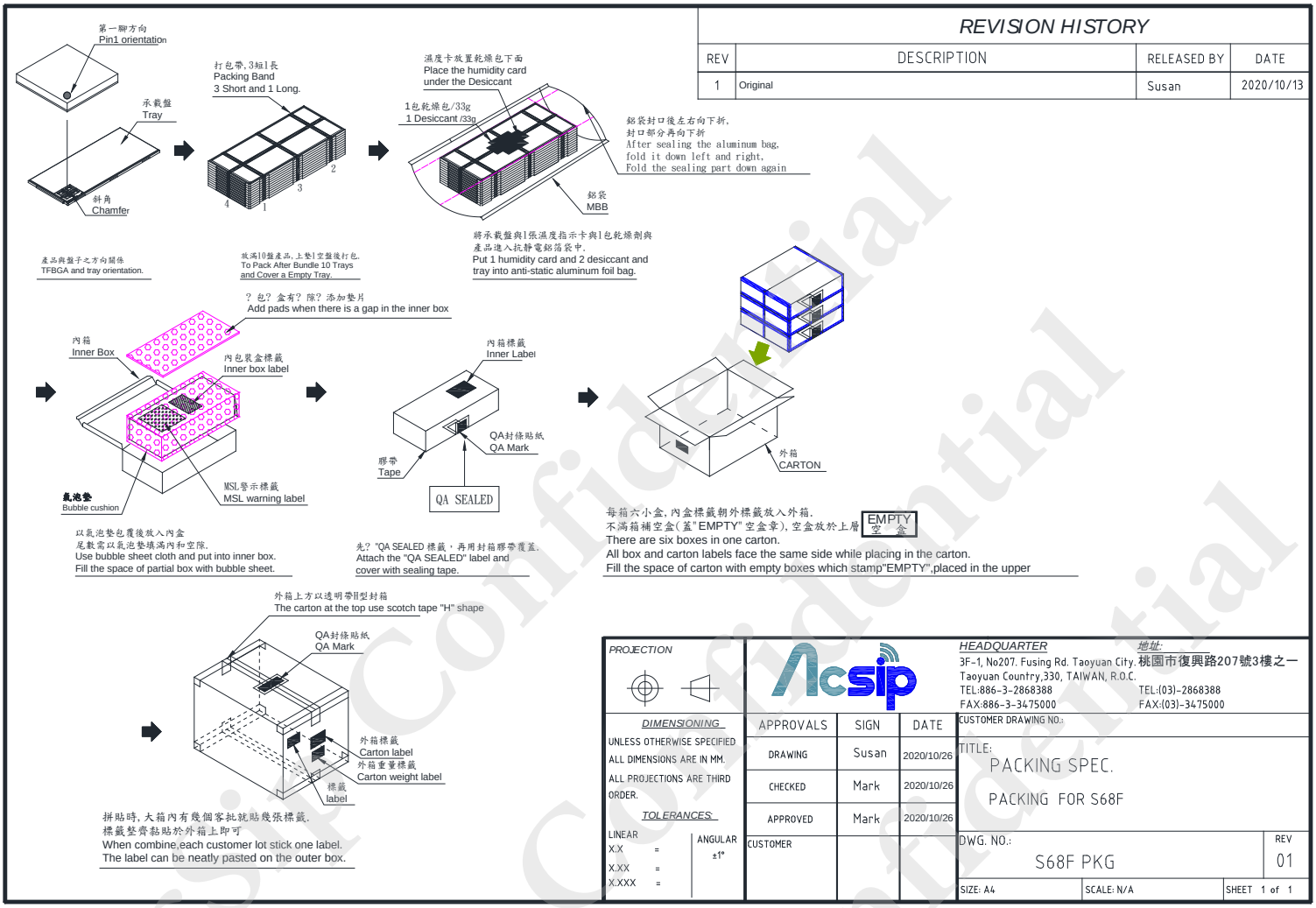
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6-2. Tray Dimension



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6-3. Packing Information



6-4. Humidity Indicator Card



Dry



Wet

Indicates：

5%, 10%, 60% relative humidity

Color Change：

Brown（Dry）→ Blue（Wet）

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